Pearls AQI Predictor: Model Training, Evaluation, and Explainability

Milestone 5 Technical Report

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***Abstract*—This report presents the training pipeline of the Pearls AQI Predictor, a serverless machine learning system forecasting the Air Quality Index (AQI) of Karachi. Historical features were read from the Hopsworks Feature Store, and three forecasting models were trained and compared: Ridge Regression, Random Forest, and a Long Short-Term Memory (LSTM) neural network. Models were evaluated using RMSE, MAE, and R-squared on a time-based hold-out set. Random Forest achieved the best performance and was registered in the Hopsworks Model Registry. SHAP was used to explain the model's predictions, confirming that it relies on physically sensible features. The analysis also demonstrates the inherent difficulty of multi-day AQI forecasting and the importance of matching model complexity to dataset size.**

***Index Terms*—AQI forecasting, Random Forest, Ridge Regression, LSTM, SHAP, Hopsworks, Model Registry**

I. Introduction

Following the exploratory data analysis of the previous milestone, this milestone focuses on building the training pipeline: reading engineered features from the Hopsworks Feature Store, training multiple forecasting models, evaluating them with standard regression metrics, selecting the best model, and registering it in the Model Registry for later use by the web application. Explainability was incorporated using SHAP, in line with the project requirements.

II. Feature Engineering for Forecasting

The forecasting task requires predicting future AQI from information available in the present. To enable this, lag features and rolling statistics were engineered from the historical AQI series, including lags at 1, 2, 3, 6, 12, 24, 48, 72, 96, 120, and 168 hours, as well as 24-hour, 72-hour, and 168-hour rolling means and a 24-hour rolling standard deviation. Recent values of key meteorological drivers (PM2.5, wind, temperature) were also lagged. Cyclical encodings of the hour and month were added so that adjacent time values are treated as numerically close. The prediction target was defined as the AQI 24 hours ahead, aligning with the project goal of forecasting upcoming days.

III. Methodology

The dataset was split chronologically rather than randomly: the earliest 80% of records were used for training and the most recent 20% for testing. This time-based split prevents information from the future leaking into training and provides a realistic estimate of forecasting performance.

Three models were trained:

- **Ridge Regression** – a regularized linear baseline (features standardized).
- **Random Forest** – a non-linear ensemble of decision trees.
- **LSTM** – a recurrent neural network implemented in PyTorch, trained on sequences of past observations.

PyTorch was used for the LSTM instead of TensorFlow due to a dependency incompatibility between the available TensorFlow build and the Hopsworks client library. The project brief permits either framework for the advanced model.

IV. Results

Table I summarizes the performance of the three models on the test set for the 24-hour-ahead forecast.

TABLE I
MODEL PERFORMANCE (24-HOUR-AHEAD FORECAST)

Model	RMSE	MAE	R^2
Ridge Regression	20.31	13.76	0.103
Random Forest	19.30	14.01	0.190
LSTM	25.84	18.76	-0.448

Random Forest achieved the best performance across RMSE and R-squared. To confirm that the model learns genuine structure, it was compared against a naive persistence baseline that predicts future AQI as equal to the current AQI; this baseline achieved an R-squared of only 0.033, which the Random Forest clearly exceeds.

Two findings are noteworthy. First, the simpler Random Forest outperformed the more complex LSTM. With a dataset of approximately 4,300 records, the LSTM overfitted the training data and failed to generalize, illustrating that greater model complexity does not guarantee better results when data is limited. Second, forecast accuracy decreases as the prediction horizon lengthens, since longer-range AQI depends increasingly on future meteorological conditions that cannot be known in advance.

V. Model Registry

The best-performing model (Random Forest), together with its feature scaler and the list of input features, was saved to the Hopsworks Model Registry. Storing the model centrally allows the automated training pipeline and the web application to retrieve and use the exact trained artifact without retraining.

VI. Explainability with SHAP

SHAP (SHapley Additive exPlanations) was applied to the Random Forest model to quantify the contribution of each feature to its predictions. The analysis identified PM2.5 as the dominant predictor, followed by the recent multi-day AQI trend (rolling averages) and seasonal information (month). The directional analysis confirmed intuitive relationships: higher PM2.5 increases predicted AQI, while higher wind speed decreases it. These results indicate that the model bases its predictions on physically meaningful factors rather than spurious correlations, supporting its trustworthiness.

VII. Conclusion

This milestone delivered a complete training pipeline that reads from the Feature Store, trains and evaluates multiple models, selects the best, and registers it for downstream use. Random Forest provided the strongest and most reliable performance, outperforming both a linear baseline and a deep learning model on this dataset. SHAP analysis confirmed the model's reliance on sensible features. The honest evaluation also highlighted the fundamental challenge of multi-day AQI forecasting. The registered model provides the foundation for the automated CI/CD pipeline and the forecasting dashboard developed in the subsequent milestones.